

From Data to Decisions: Optimizing Pediatric Vaccine Tender Schedules under Uncertainty

Doctoral Dissertation Proposal

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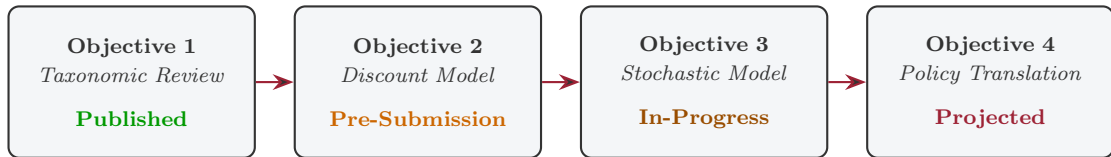
Outline

- 1 Introduction
- 2 Objective 1 - Taxonomic Review
- 3 Objective 2 - Discount Model
- 4 Objective 3 - Stochastic Extension
- 5 Objective 4 - Policy Translation
- 6 Proposed Research Timeline

Introduction & Research Motivation

- **Global Public Health Impact:** Pediatric immunization campaigns prevent an estimated 3.5 to 5 million deaths annually from vaccine-preventable diseases [12].
- **Fragility of Procurement Systems:** Low- and Middle-Income Countries (LIC/LMICs) rely heavily on centralized, pooled purchasing and funding frameworks managed by global intermediaries (UNICEF, Gavi, PAHO) [6, 15].
- **The Core Mechanism:** The *vaccine tender* is a multi-year planning framework through which producers commit specific volumes at negotiated tiered price points.
- **The Current Operational Gap:**
 - ▶ Intermediaries rely on reactive, short-term rolling schedules rather than 5-to-10-year strategic planning horizons.
 - ▶ Multi-antigen dynamics (e.g., substitution between standalone antigens and complex combinations like pentavalent/hexavalent formulations) are handled in isolation.

Program of Research: Methodological Framework



[Objective 1 - Taxonomic Review]
[Complete]

Completed Contribution: Literature Taxonomy (Obj 1)

● Review Scope

- ▶ 138 publications identified across procurement, pricing, supply chain, scheduling, and vaccine policy literature.
- ▶ Developed a four-dimensional taxonomy to classify vaccine tender and procurement research.

● Search Parameters

- ▶ *vaccine tender procurement, pediatric vaccine tendering, vaccine tender scheduling*

Search Categories

Vaccine Tender	Scheduling, Contracting, Process, Procurement
Scheduling	Vaccine Tender, Vaccine Tender Procurement, Pediatric Immunizations
Procurement	Long-Term Vaccine Tender, Vaccine Tender Scheduling
Pediatric	Vaccine Tender, Vaccine Procurement, Vaccine Tender Scheduling, Vaccine Purchasing
Vaccine	Tender Market Design, Market Design, Procurement

Completed Contribution: Literature Taxonomy (Obj 1)

● Limited Comprehensive Research

- ▶ Few studies jointly address vaccine tender scheduling and long-term procurement.
- ▶ Existing research emphasizes procurement decisions, while long-term tender scheduling remains largely unexplored.
- ▶ Scheduling and procurement are typically treated as independent decisions rather than coordinated processes.

● Lack of OR Methodology

- ▶ Optimization models exist for vaccine bundling, formularies, and supply chains.
- ▶ Comparable models for vaccine tender scheduling and procurement are largely absent.
- ▶ Limited application of stochastic optimization, game theory, and AI/ML methods within this domain.

● Future Research Directions

- ▶ All unit quantity discounts.
- ▶ Stochastic optimization for uncertain vaccine demand and supply.
- ▶ Coordinated multi-vaccine and multi-producer tender planning.

Dissertation Motivation

Literature Gap

No identified framework jointly addresses

- ➊ Long-term tender scheduling,
- ➋ Procurement planning,
- ➌ Multi-vaccine coordination,
- ➍ Demand uncertainty.

Research Response

Objectives 2–4 develop a stochastic optimization framework for pediatric vaccine tender scheduling and procurement.

[Objective 2 - Discount Model]
[Advisor Review]

Discount Model (Obj 2)

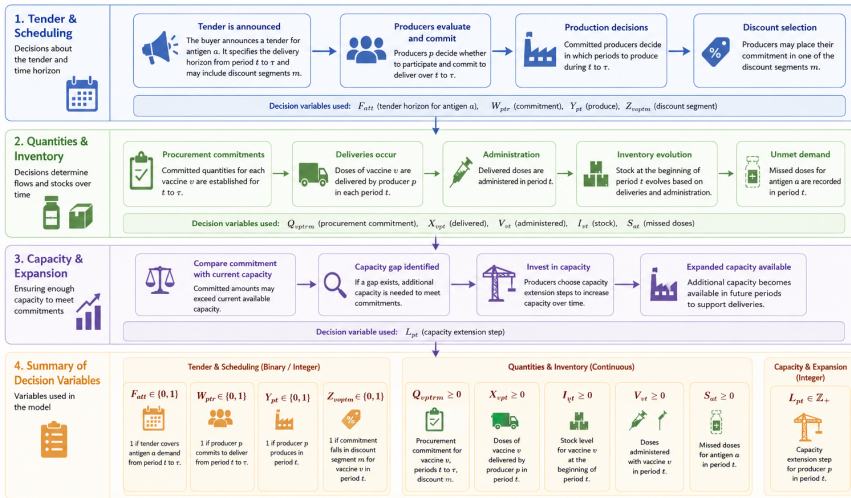
- **Problem Setting:** Develop a strategic vaccine scheduling and procurement framework with quantity discounts for the Hypothetically Coordinated Vaccine Market (HCVM).
- **Key Modeling Features:**
 - ① Flexible tender lengths of varying durations rather than fixed re-tendering cycles.
 - ② Quantity-based discount schedules representing alternative buyer- and producer-side pricing incentives.
 - ③ Multi-producer participation decisions that include capacity increase decisions, and capture competition, market entry, and market concentration effects.
 - ④ Multi-vaccine procurement planning across a portfolio of substitute and complementary vaccines.
- **Model Objectives:**
 - ▶ Evaluate procurement policies under alternative stakeholder perspectives:
 - ★ UNICEF-GAVI (cost and social benefit balance)
 - ★ Social Benefit Maximization
 - ★ producer Profit Maximization
 - ▶ Measure impacts on procurement cost, producer profitability, market participation, schedule complexity, and missed doses.

Discount Model (Obj 2)

- **Problem Setting:** Develop a strategic vaccine scheduling and procurement framework with quantity discounts for the Hypothetically Coordinated Vaccine Market (HCVM).
- **Research Significance:**
 - ▶ Extends vaccine tender planning from isolated contract decisions to coordinated long-term procurement strategies.
 - ▶ Quantifies how discount structures influence producer behavior, market concentration, and vaccine availability.
 - ▶ Provides evidence-based guidance for designing procurement mechanisms that improve coverage while maintaining supply chain resilience.

Tender Scheduling Framework (Obj 2)

From Tender to Delivery: Process Leading to the Need for Decision Variables



Why these variables? Producers may commit more than their current capacity because they can increase capacity in time. The variables above capture the entire decision process—from the tender and commitment choices, through deliveries and inventory, to capacity investment—needed to meet demand.

Decision Variables

Tender & Scheduling (Binary/Integer)

- $F_{at\tau} \in \{0, 1\}$: 1 if tender covers antigen a demand from period t to τ .
- $W_{pt\tau} \in \{0, 1\}$: 1 if producer p commits to deliver from period t to τ .
- $Y_{pt} \in \{0, 1\}$: 1 if producer p produces in period t .
- $Z_{vptm} \in \{0, 1\}$: 1 if commitment falls in discount segment m for vaccine v in period t .
- $L_{pt} \in \mathbb{Z}_+$: Capacity extension step for producer p in period t .

Quantities & Inventory (Continuous)

- $Q_{vpt\tau m} \geq 0$: Procurement commitment for vaccine v , periods t to τ , discount m .
- $X_{vpt} \geq 0$: Doses of vaccine v delivered by producer p in period t .
- $I_{vt} \geq 0$: Stock level for vaccine v at the beginning of period t .
- $V_{vt} \geq 0$: Doses administered with vaccine v in period t .
- $S_{at} \geq 0$: Missed doses for antigen a in period t .

Discount Model Formulation

[UG] Balance Objective Function

$$\min \sum_{t \in \mathcal{T}} \delta_t \left(\sum_{a \in \mathcal{A}} \sum_{(t, \tau) \in \mathcal{F}} g_t F_{at\tau} + \sum_{v \in \mathcal{V}} \sum_{p \in \mathcal{P}} \sum_{m \in \mathcal{M}} \bar{r}_{vt} (1 - \zeta_{vm}) \sum_{(t, \tau) \in \mathcal{F}} Q_{vpt\tau m} + \sum_{a \in \mathcal{A}} \beta S_{at} + \sum_{v \in \mathcal{V}} h_v \bar{r}_{vt} I_{vt} \right)$$

Key Indices & Sets:

- $\mathcal{A}, \mathcal{V}, \mathcal{P}$: Antigens, Vaccines, and Producers.
- $\mathcal{M}$: Discrete quantity discount segments.
- $\mathcal{F}$: the set of all valid tender index pairs (t, τ) , where t is the start period and τ is the end period, and each tender lasts at most Δ periods, i.e.,

$$\mathcal{F} = \{(t, \tau) : t \in \mathcal{T}, t \leq \tau \leq \min\{t + \Delta, |\mathcal{T}|\}\}.$$

Cost Term Decompositions:

- Term 1: Administrative tender setup costs (g_t).
- Term 2: Variable procurement costs under discount ζ_{vm} .
- Term 3: Missed dose penalty (β)
- Term 4: Inventory holding costs (h_{vt}).

Discount Model Formulation

[SB] Social Benefit Objective Function

$$\min \sum_{t \in \mathcal{T}} \sum_{a \in \mathcal{A}} S_{at}$$

Key Indices & Sets:

- $\mathcal{A}$: Antigens.

Cost Term Decompositions:

- None

**SB model pending re-runs with ROI constraint*

Discount Model Formulation

[MP] Max Profit Objective Function

$$\min \sum_{t \in \mathcal{T}} \delta_t \left(\sum_{v \in \mathcal{V}_p} \sum_{p \in \mathcal{P}} \sum_{m \in \mathcal{M}} -\bar{r}_{vt}(1 - \zeta_{vm}) \sum_{(t, \tau) \in \mathcal{F}} Q_{vpt\tau m} + \sum_{p \in \mathcal{P}} \gamma_p L_{pt} + \sum_{v \in V_p} \sum_{p \in \mathcal{P}} \sum_{(t, \tau) \in \mathcal{F}} f_{vpt\tau} W_{pt\tau} \right)$$

Key Indices & Sets:

- $\mathcal{A}$, $\mathcal{V}$, $\mathcal{P}$: Antigens, Vaccines, and Producers.
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Cost Term Decompositions:

- Term 1: Variable procurement costs under discount ζ_{vm} .
- Term 2: Production capacity increase cost (γ_p)
- Term 3: Proportion of production set-up cost recoupment $f_{vpt\tau}$

Tender Scheduling Constraints

Tender Scheduling

$$\sum_{\tau: (t, \tau) \in \mathcal{F}} F_{at\tau} \leq 1, \quad \forall a \in \mathcal{A}; t \in \mathcal{T}$$

$$\sum_{\substack{\max\{1, l-\Delta\} \leq t \leq l \\ (t, \tau) \in \mathcal{F}}} F_{at\tau} \geq 1, \quad \forall a \in \mathcal{A}, l \in \mathcal{T}$$

Ensures continuous scheduling while preventing overlapping tender start periods.

Tender Participation

$$(\tau - t + 1)F_{at\tau} \leq \sum_{p \in \mathcal{P}_a} \sum_{l=t}^{\tau} Y_{pl}, \quad \forall a \in \mathcal{A}; (t, \tau) \in \mathcal{F}$$

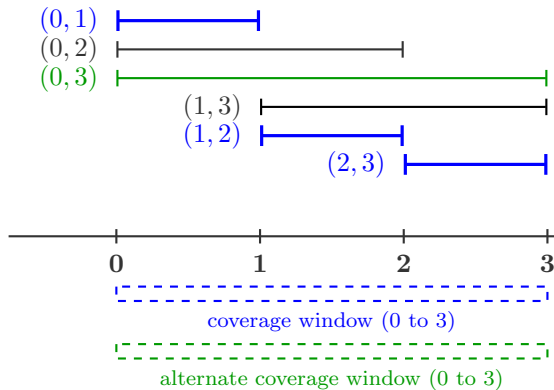
$$\sum_{a \in \mathcal{A}_p} F_{at\tau} \geq W_{pt\tau}, \quad \forall p \in \mathcal{P}; (t, \tau) \in \mathcal{F}$$

$$\sum_{a \in \mathcal{A}_p} F_{at\tau} \leq |\mathcal{A}_p| W_{pt\tau}, \quad \forall p \in \mathcal{P}; (t, \tau) \in \mathcal{F}$$

Links participation decisions (Y) with tender commitments (W) through feasibility coupling.

Tender Coverage Logic (Illustrative Example: $T = \{1, 2, 3\}$)

Feasible $F_{at\tau}$ (intervals)



Constraint 1

← --- At most one start per period t

Constraint 2

Each period l must be covered by at least one active interval in window

Commitment and Discount Constraints

Tender Commitments

$$\sum_{v \in \mathcal{V}_p} \sum_{m \in \mathcal{M}} Q_{vpt\tau m} \leq W_{pt\tau} \left(\sum_{l=t}^{\tau} \left[\bar{s}_p + \sum_{k=1}^l \kappa(\tau - t + 1) \bar{s}_p L_{pk} \right] \right), \quad \forall p \in \mathcal{P}; (t, \tau) \in \mathcal{F}$$

$$\sum_{v \in \mathcal{V}_p} \sum_{m \in \mathcal{M}} \bar{r}_{vt}(1 - \zeta_{vm}) Q_{vpt\tau m} \geq \sum_{v \in \mathcal{V}_p} (1 + l_{vp}) f_{vpt\tau} W_{pt\tau}, \quad \forall p \in \mathcal{P}; (t, \tau) \in \mathcal{F}$$

Commitments remain within available capacity while generating sufficient return to partially recover setup investments.

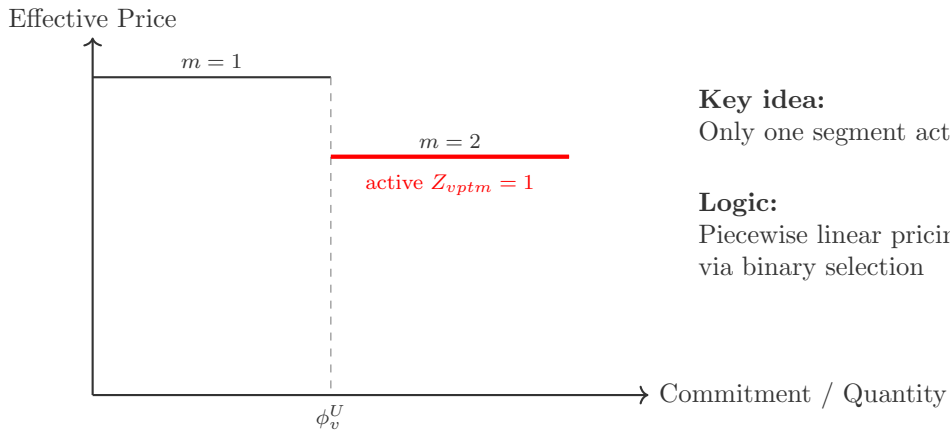
Discounts

$$\phi_{vm}^L Z_{vptm} \leq \sum_{(t, \tau) \in \mathcal{F}} Q_{vpt\tau m} \qquad \sum_{m \in \mathcal{M}} Z_{vptm} \leq 1$$

$$\phi_{vm}^U Z_{vptm} \geq \sum_{(t, \tau) \in \mathcal{F}} Q_{vpt\tau m}$$

All-unit quantity discounts with mutually exclusive discount tiers across vaccines, producers and time periods.

Manufacturer Discount Schemes (2) (Piecewise Structure)



Key idea:

Only one segment active

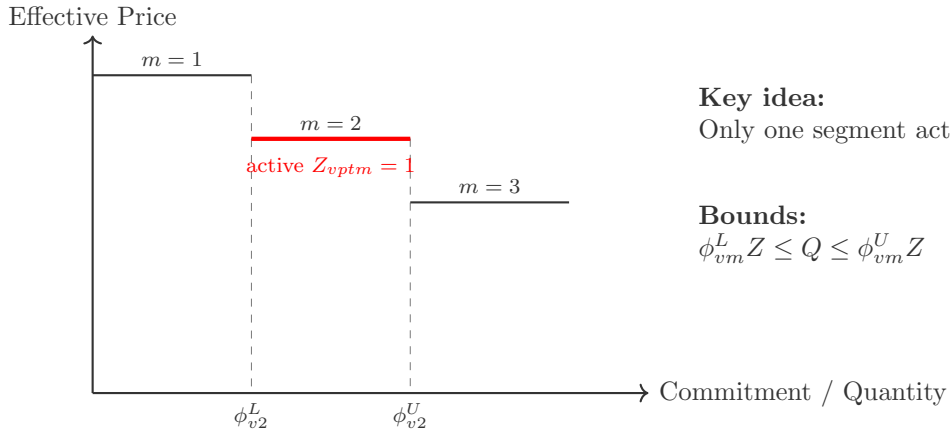
Logic:

Piecewise linear pricing
via binary selection

$$\phi_{vm}^L Z_{vptm} \leq \sum_{(t,\tau) \in \mathcal{F}} Q_{vpt\tau m} \leq \phi_{vm}^U Z_{vptm}$$

$$\sum_{m \in \{1,2\}} Z_{vptm} \leq 1$$

Manufacturer Discount Schemes (3) (Piecewise Structure)



Key idea:

Only one segment active

Bounds:

$$\phi_{vm}^L Z \leq Q \leq \phi_{vm}^U Z$$

$$\phi_{vm}^L Z_{vptm} \leq \sum_{(t,\tau) \in \mathcal{F}} Q_{vpt\tau m} \leq \phi_{vm}^U Z_{vptm}$$

$$\sum_{m \in \mathcal{M}} Z_{vptm} \leq 1$$

Delivery and Balance Constraints

Vaccine Deliveries

$$\sum_{m \in \mathcal{M}} Q_{vpt\tau m} \geq W_{pt\tau} \sum_{l=t}^{\tau} X_{vpl}, \quad \forall v \in \mathcal{V}, p \in \mathcal{P}_v, (t, \tau) \in \mathcal{F}$$

$$\sum_{v \in \mathcal{V}_p} X_{vpt} \leq \bar{Y}_{pt} \left(s_{pt} + \sum_{l=1}^t \kappa \bar{s}_p L_{pl} \right), \quad \forall p \in \mathcal{P}, t \in \mathcal{T}$$

Ensures delivered vaccines (X) do not exceed committed quantities (Q) or available capacity.

Inventory and Balance

$$I_{v,t-1} + \sum_{p \in \mathcal{P}_v} X_{vpt} = V_{vt} + I_{vt}, \quad \forall v \in \mathcal{V}, t \in \mathcal{T}$$

$$d_{at} - \sum_{v \in \mathcal{V}_a} V_{vt} + S_{a,t-1} \leq S_{at}, \quad \forall a \in \mathcal{A}, t \in \mathcal{T}$$

Balances inventory flows, vaccine production, and unmet demand propagation over time.

Computational Performance Profile

Experiments conducted on **30 independent demand portfolios** generated via moving average forecasting architectures using actual data from the Linksbridge Vaccine Almanac [10].

Table: Mean solve time (seconds) and standard deviation across 30 trials by objective/discount structure

Problem Specification	UG		SB		MP	
	Mean	SD	Mean	SD	Mean	SD
1: 1 segment	53.27	28.30	21.50	10.42	40.03	25.67
2: 2 segments	485.41	227.42	132.70	49.83	355.05	239.59
3: 2 segments, penta/hexa discount	1,746.65	2,940.63	180.77	61.10	437.66	98.75
4: 2 segments, no penta/hexa	417.10	85.68	156.40	35.72	1,708.30	1,736.39
5: 3 segments	229.93	80.86	75.20	33.56	170.01	89.92

- **Complexity Driver:** High computational complexity driven by a vast solution space of multi-producer, multi-vaccine, and tiered-pricing combinations, further intensified by unpredictable demand.

Strategic Analysis of Preliminary Results

- **Objective Dominance Dynamic:** Total system expense and scheduling fragmentation are functions of the institutional procurement strategy. The commercial orientation ([MP]) results in highly fractured, short-term tender structures.
- **The Multivalent Safety Net:** Sensitivity analysis isolating single-antigen supply chains shows that removing combination formulations (pentavalent/hexavalent) causes an exponential surge in missed doses. The system lacks the base capacity to effectively cover the loss, given the current set of less-valent vaccines.

Market Concentration vs. Redundancy Risk

All-unit volume discounts incentivize procurement agents to bundle awards with large suppliers, effectively decreasing competition for smaller producers, and increasing supply chain risk.

Objective Dominance Dynamic

Objective Dominance Dynamic

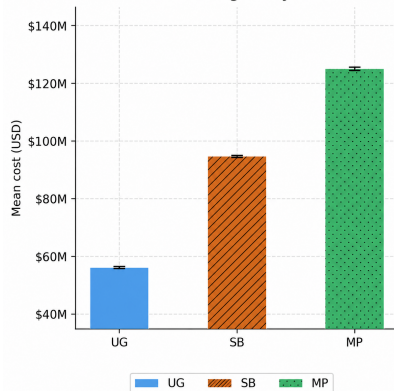
Total system expense and scheduling fragmentation are functions of the institutional procurement strategy.

Overlapping Tender Periods by Disease — Mean $\pm$ 95% CI (30 Trials)

Disease	UG	SB	MP
Diphtheria	2.97 $\pm$ 0.63	17.87 $\pm$ 2.66	20.63 $\pm$ 2.00
HPV	1.33 $\pm$ 0.30	3.90 $\pm$ 1.47	65.47 $\pm$ 2.08
Hepatitis B	3.90 $\pm$ 0.78	15.73 $\pm$ 2.21	15.57 $\pm$ 2.66
Hib	5.43 $\pm$ 0.91	17.80 $\pm$ 2.27	14.77 $\pm$ 2.39
PCV	0.60 $\pm$ 0.20	3.27 $\pm$ 1.00	38.90 $\pm$ 1.58
Pertussis	4.47 $\pm$ 0.68	18.97 $\pm$ 2.22	13.63 $\pm$ 2.11
Polio	0.00 $\pm$ 0.00	0.03 $\pm$ 0.06	0.40 $\pm$ 0.47
Rotavirus	0.03 $\pm$ 0.06	0.20 $\pm$ 0.14	3.37 $\pm$ 1.35
Tetanus	3.17 $\pm$ 0.57	17.70 $\pm$ 2.60	21.87 $\pm$ 2.13

Values represent mean number of overlapping tender periods $\pm$ 95% CI.

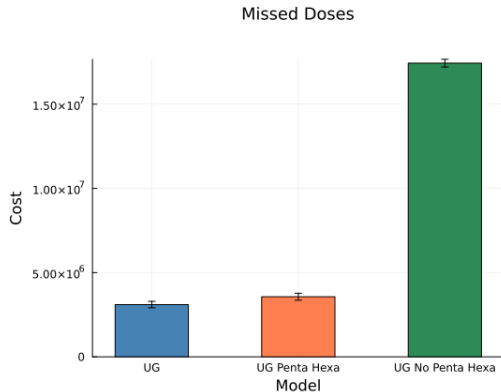
Coordinating entity costs



The Multivalent Safety Net

The Multivalent Safety Net

Sensitivity analysis isolating single-antigen supply chains shows that removing combination formulations (pentavalent/hexavalent) causes an exponential surge in missed doses. The system lacks the base capacity to effectively cover the loss, given the current set of less-valent vaccines.

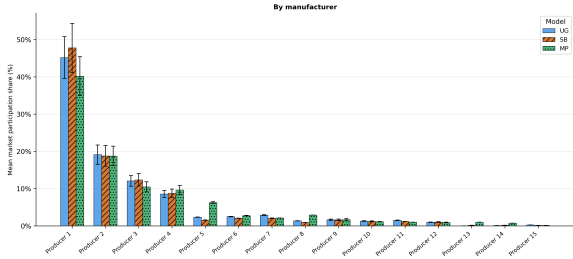


Market Concentration vs. Redundancy Risk

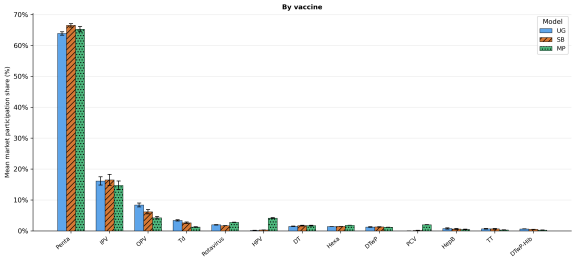
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All-unit volume discounts incentivize procurement agents to bundle awards with large suppliers, effectively decreasing competition for smaller producers, and increasing supply chain risk.

Market Participation Across 30 Trials — Model Comparison
Mean share $\pm$ 95% CI (aggregated by model family)



Market Participation Across 30 Trials — Model Comparison
Mean share $\pm$ 95% CI (aggregated by model family)



[Objective 3 - Stochastic Extension]
[In-Progress]

Potential Research Questions for Paper 2

Proposed Two-Stage Stochastic Framework

- **Demand–Supply Interaction in Procurement Systems:** How do stochastic demand and supply conditions interact with procurement design and market structure to shape vaccine availability, missed doses, and supplier participation?
- **Multi-Period Optimization Under Uncertainty:** How should multi-period procurement and allocation decisions be optimized under uncertainty to balance resilience, affordability, and supplier incentives?
- **Objective Functions and Pricing Mechanisms:** how do pricing mechanisms impact market participation?
- **Value of Forecasting Accuracy:** What is the value of improved demand and supply forecasting accuracy in procurement planning and contract design?

Projected Work: Two-Stage Stochastic Recourse (Obj 3)

Balance [UG] Model First Stage

$$\min \sum_{t \in \mathcal{T}} \delta_t \left(\sum_{a \in \mathcal{A}} \sum_{(t, \tau) \in \mathcal{F}} g_t F_{at\tau} + \sum_{v \in \mathcal{V}_{\mathcal{P}}} \sum_{p \in \mathcal{P}} \sum_{m \in \mathcal{M}} \bar{r}_{vt} (1 - \zeta_{vm}) \sum_{(t, \tau) \in \mathcal{F}} Q_{vpt\tau m} \right) + \mathbb{E}_{\Omega}[\mathcal{Q}(\bar{\Xi}, \omega)]$$

Balance [UG] Model Second Stage

$$\mathcal{Q}(\bar{\Xi}, \omega) = \min \sum_{t \in \mathcal{T}} \delta_t \left(\sum_{a \in \mathcal{A}} \beta S_{at\omega} + \sum_{v \in \mathcal{V}} h_v \bar{r}_{vt} I_{vt\omega} \right)$$

Projected Work: Two-Stage Stochastic Recourse (Obj 3)

Social Benefit [SB] Model First Stage

$$\min \mathbb{E}_{\Omega}[\mathcal{Q}(\bar{\Xi}, \omega)]$$

Social Benefit [SB] Model Second Stage

$$\mathcal{Q}(\bar{\Xi}, \omega) = \min \sum_{t \in \mathcal{T}} \sum_{a \in \mathcal{A}} S_{at\omega}$$

Projected Work: Two-Stage Stochastic Recourse (Obj 3)

Max Profit [MP] Model Objective First Stage

$$\min \sum_{t \in \mathcal{T}} \delta_t \left(\sum_{v \in \mathcal{V}_p} \sum_{p \in \mathcal{P}} \sum_{m \in \mathcal{M}} -\bar{r}_{vt}(1 - \zeta_{vm}) \sum_{(t, \tau) \in \mathcal{F}} Q_{vpt\tau m} + \sum_{p \in \mathcal{P}} \gamma_p L_{pt} + \sum_{v \in \mathcal{V}_p} \sum_{p \in \mathcal{P}} \sum_{(t, \tau) \in \mathcal{F}} f_{vpt\tau} W_{pt\tau} \right)$$

Max Profit [MP] Model (Dual) Second Stage

$$\max \sum_{v, p, t, \tau} \alpha_{vpt\tau}^\omega \sum_m \bar{Q}_{vpt\tau m} - \sum_{p, t} \pi_{pt}^\omega \bar{Y}_{pt} \left(s_{pt} + \sum_{l=1}^t \kappa \bar{s}_p \bar{L}_{pl} \right) - \sum_{a, t} \nu_{at}^\omega d_{at}^\omega + \sum_v \mu_v^\omega \hat{I}_{v0}$$

- **Objective Function:** The second stage contains no recourse variables. Instead, it functions purely as a feasibility check and cut-generating problem.

Projected Work: Two-Stage Stochastic Extension (OBJ 3.b)

Chance Constraint

$$P\left(\tilde{d}_{at} \leq \sum_{v \in \mathcal{V}_a} V_{vt}\right) \geq (1 - \epsilon), \quad \forall a \in \mathcal{A}; t \in \mathcal{T}$$

- **Chance Constraint:** Specifically, let ϵ denote the allowable proportion of vaccine demand that may go unmet. The resulting coverage requirement is therefore $(1 - \epsilon)$, which is set to 90% in alignment with UNICEF vaccination coverage goals, corresponding to $\epsilon = 0.10$ [14].

Solution Approach: Stabilized Benders Decomposition

- **Decomposition Architecture:** We solve the stochastic program using a stabilized Benders (L-shaped) decomposition that separates first-stage procurement decisions from second-stage operational responses under uncertainty.
- **Master Problem Role:** The MP optimizes procurement, scheduling, and allocation decisions while incorporating an auxiliary variable (θ) that approximates expected second-stage cost. It is a MILP containing all binary and integer decision variables.
- **Subproblem Role:** The SP evaluates feasibility and cost of first-stage decisions across stochastic realizations of demand and supply. It provides scenario-dependent feedback on system performance under uncertainty.
- **Iterative Refinement:** Dual information from the SP is used to generate optimality (and feasibility) cuts, progressively tightening the MP approximation until convergence of LB and UB.

Solution Approach: The Problem

Cut-Based Information Feedback Loop

The decomposition framework iteratively transfers dual information from scenario-based subproblems into linear cuts that refine the master problem's approximation of second-stage recourse costs, enabling convergence of lower and upper bounds over successive iterations.

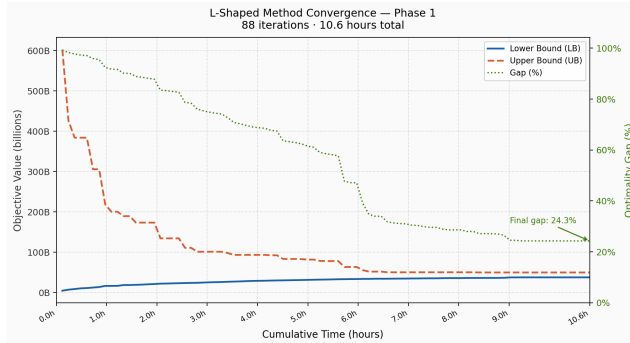


Figure: Stochastic Balance Model Convergence: 30 scenarios

Solution Approach: The Solution

Cut-Based Information Feedback Loop

The decomposition framework iteratively transfers dual information from scenario-based subproblems into linear cuts that refine the master problem's approximation of second-stage recourse costs, enabling convergence of lower and upper bounds over successive iterations.

- **Acceleration Strategy:** Investigate multiple enhancement mechanisms to improve convergence speed and numerical stability of the L-shaped method.
- ➊ Multi-cut generation (optimality) [2, 1]
 - ➋ Partial Benders [4, 5, 3]
 - ➌ Pareto-optimal cuts (Magnanti–Wong cuts) [11]
 - ➍ Regularization (trust region) [13, 7, 9]
 - ➎ Parallel L-shaped method [9]
 - ➏ Level-set methods (ϵ -optimality) [8]

[Objective 4 - Policy Translation]
[Complete]

Projected Work: Stochastic Policy Analysis (Obj 4)

- **Demand-Responsive Tendering:** Evaluate whether tender lengths should adapt to forecast uncertainty, and how uncertainty affects market conditions and participation.
- **Risk-Based Discount Design:** Examine whether quantity discount schedules should vary according to demand uncertainty, balancing procurement savings against the risk of over- or under-commitment.
- **Strategic Procurement Flexibility:** Determine when planners should prioritize procurement adaptability over immediate cost savings, particularly in markets characterized by high demand uncertainty.
- **Supplier Diversification Policies:** Quantify the trade-off between short-term cost reductions from market concentration and the long-term resilience benefits of maintaining multiple qualified suppliers.
- **Scenario-Driven Procurement Rules:** Develop practical planning guidelines that map observable market conditions (demand variability, supplier availability, and vaccine portfolio composition) to recommended tendering strategies.

Proposed Research Timeline

Jun – Aug 2026

Objective 2

Final Paper
Revisions &
Submission

Sep – Nov 2026

Objective 3

Algorithmic
Coding & Scenario
Experiments

Dec – Feb 2027

Objective 4

Analysis & Policy
Translation
Framework

Mar – Apr 2027

Dissertation

Assembly, Defense
Prep, & Final
Oral Defense

Target Graduation Window: End Spring 2027

Questions/Discussion

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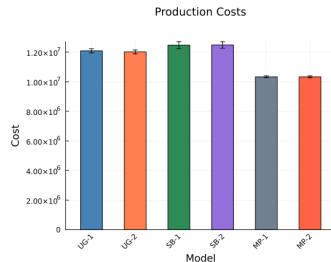
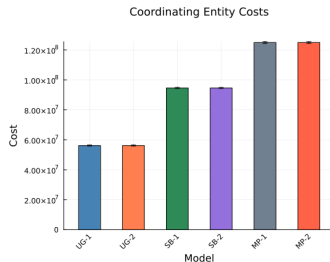
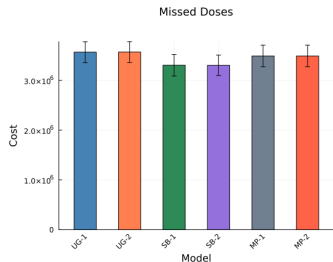
Global socioeconomic inequalities in vaccination coverage, supply, and confidence.
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BACKUP SLIDES

Buyer v Producer Discounts

Buyer v Producer discounts

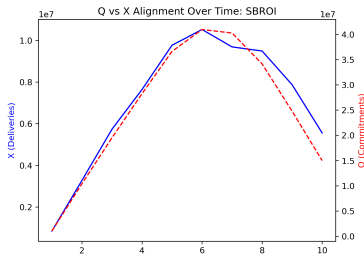
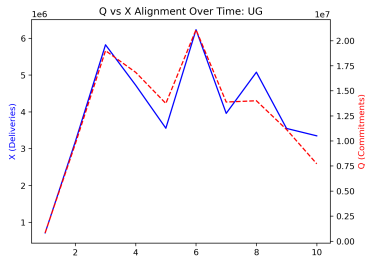
Buyer v Producer discounts are statistically insignificant, and when considered from the UNICEF/GAVI perspective, are unfavorable to buyer discounts.



Commitment (Q) v Deliveries (X)

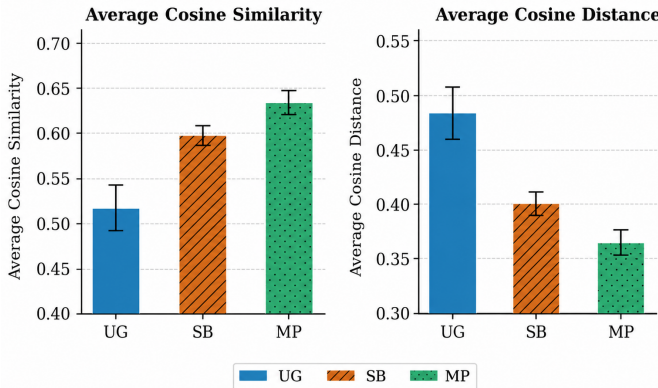
Commitment versus deliveries

Commitments generally mirror deliveries, though increase slightly across model objectives.



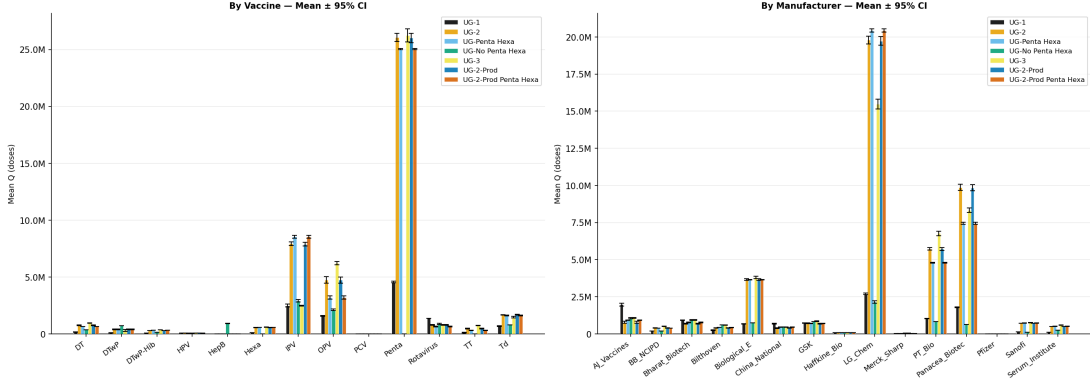
Cosine Similarity and Distance

Distance and Similarity Metrics — Mean $\pm$ 95% CI across 30 Trials



Market Participation

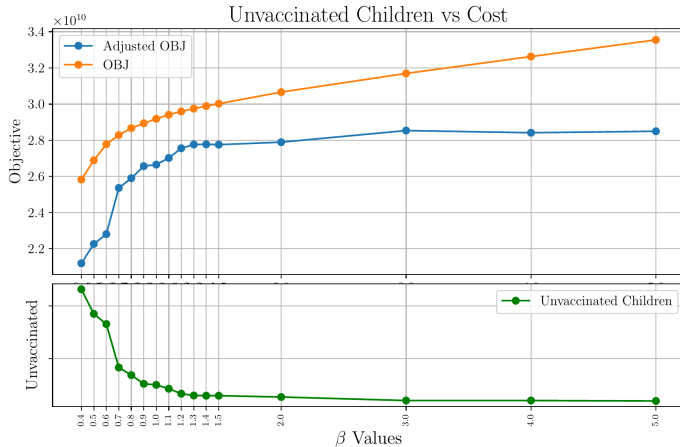
Total Q Across 30 Trials — Model Comparison
Mean $\pm$ 95% CI



Pareto Frontier - Beta

Missed Dose Penalty

Beta must take a value greater than the average cost per dose of all vaccines



Pareto Frontier - Discounts

Commitment versus deliveries

2 segment example. Discount breakpoint selected where 57% in commitments is seen.

